Google Skills

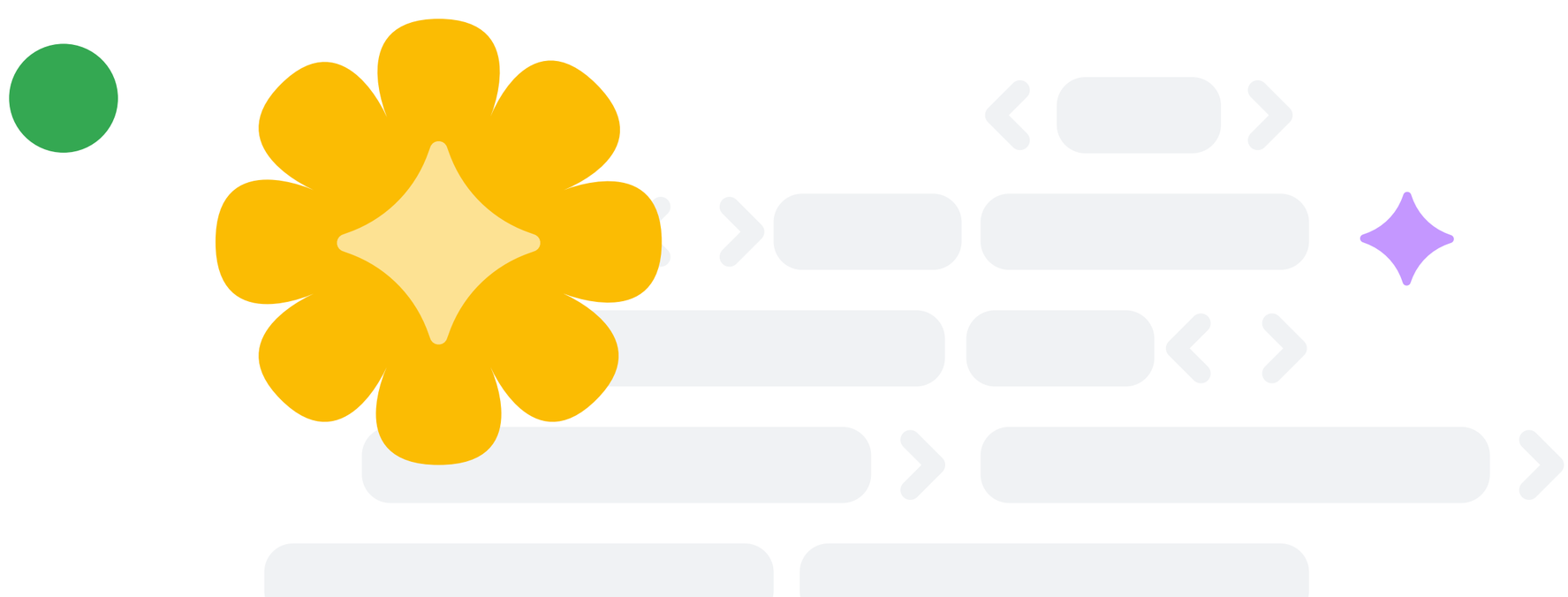
The solution: Agent abstraction

Why agents are different

An agent isn’t just an LLM with tools bolted on. It’s a fundamentally different abstraction; a system that can pursue goals autonomously. Instead of responding to each request in isolation, an agent maintains context, plans ahead, and takes initiative to achieve objectives.

According to Google's agent whitepaper, in its most fundamental form:

“A generative AI agent can be defined as an application that attempts to achieve a goal by observing the world and acting upon it using the tools that it has at its disposal. Agents are autonomous and can act independently of human intervention, especially when provided with proper goals or objectives they are meant to achieve. Agents can also be proactive in their approach to reaching their goals.”



Not everything that uses an LLM and tools is an agent. Breaking down this definition, true agents are characterized by five essential capabilities:

1 Goal-directed behavior

Agents work toward objectives, not just respond to queries. They understand the difference between the current state and the desired state, and they take actions to bridge that gap,

2 Autonomous operation

Agents can act independently of human intervention. Once given a goal, they determine the steps needed without requiring explicit instruction for each action.

3 Proactive initiative

Agents don't just respond—they actively work toward goals. Even in the absence of explicit instruction sets, an agent can reason about what it should do next to achieve its ultimate goal.

4 Environmental awareness

Agents observe and perceive their environment through various inputs like user requests, API responses, database states, and error messages. They build a model of the world and update it as they work.

5 Tool use

Agents interact with the outside world using the tools at their disposal. This might include APIs, databases, external services, or other systems, enabling them to go beyond just generating text to actually performing actions.

Seeing these characteristics at work

Let's see how these characteristics manifest with a simple example: checking the weather and deciding what to wear.

With an LLM:

```
You:   "Should I wear a jacket
       today?"
LLM:   "I don't have access to
       current weather data, but
       generally you should wear a
       jacket if it's below 60°F or
       raining."
```

The LLM has knowledge but no connection to reality.

- ✗ No environmental awareness
- ✗ No tool use

With LLM + function calling:

```
You:   "Check the weather for
       Seattle"
LLM:   [Calls weather API]
       Result: "It's 55°F and cloudy
       in Seattle"
You:   "Should I wear a jacket?"
LLM:   "Yes, at 55°F you should wear
       a jacket"
```

Better—it can use tools, but you're managing the conversation flow.

- ✓ Tool use
- ✗ Not autonomous or proactive

With an agent:

```
You:    "Should I wear a jacket
        today?"
Agent:  [Checks your location]
        [Queries weather API]
        [Checks your calendar for
        outdoor activities]
        [Considers the forecast for
        when you'll be outside]
        "Yes, definitely wear a
        jacket. It's 55°F now, but it
        will drop to 48°F by 6 PM
        when you have your outdoor
        meeting."
```

The agent autonomously gathered context and provided a complete answer.

- ✓ All five characteristics in action: goal-directed, autonomous, proactive, environmentally aware, and using tools.

Core concepts

Comparing the three approaches

Let's visualize how the same request gets handled differently across LLMs, function calling, and agents:

```
graph TD
    Q[Book a trip to Paris]

    Q --> LLM[LLM Response]
    Q --> FC[Function Calling]
    Q --> Agent[Agent Response]

    LLM --> L1[Advice Only:<br/>How to book flights...<br/>Step 1: Go to website...<br/>Step 2: Search dates...]
    FC --> F1[Needs Direction:<br/>I can search flights<br/>What dates?<br/>What airline?]
    Agent --> A1[Ready to Act:<br/>✓ Found 3 flight options<br/>✓ Matched with hotels<br/>✓ Ready to book]

    style LLM fill:#fcc
    style FC fill:#ffc
    style Agent fill:#cfc
```

Figure: The same request, three different responses—showing the power of agent autonomy.

The problems only agents solve

Some problems are uniquely suited to be solved by agents, as they require these five characteristics working in concert:

Complex multi-step workflows:
Consider expense report processing. An agent can extract receipts from emails, categorize expenses, check against policy, route for approval, and update accounting systems—all from a single request. This requires goal-directed behavior, autonomous operation and tool use working in concert.

Dynamic problem solving:
When booking travel, an agent can handle the reality that flights sell out, prices change, and schedules conflict. It adapts its approach based on what it discovers through environmental awareness and proactive initiative.

Ongoing task management:
Agents can monitor situations and take action when needed. “Keep track of flight prices to Tokyo and book when they drop below $800” isn’t a one-time query—it’s an ongoing responsibility that requires all five characteristics to function effectively.

Integrated experiences:
Instead of using five different tools to accomplish one goal, agents can coordinate across multiple systems seamlessly. Planning an event might involve calendar systems, venue databases, catering services, and invitation platforms—an agent can orchestrate all of these autonomously using the appropriate tools.

Key takeaways

The agent abstraction represents a fundamental shift in how we think about AI systems:

- **From reactive to proactive:** Agents don’t just respond; they pursue goals.
- **From isolated to integrated:** Agents maintain context and state across interactions.
- **From advisory to executive:** Agents don’t just tell you what to do; they do it.
- **From static to adaptive:** Agents adjust their approach based on what they learn.

This isn’t just an incremental improvement, but a paradigm shift enabling entirely new application categories.